

HE2003 Econometrics I

Quiz Practice 1 (Questions)

AY 2026/2027, Semester 1

Quantitative Research Society @NTU

September 28, 2026

Overview

- **Coverage:** Lectures 1–5 and Lecture 6 slides 1–3 (large-sample distribution of $\hat{\beta}_1$ and its standard error); no t-tests, confidence intervals, p-values or Stata.
- **Format:** 5 questions, 50 marks total, approximately 60 minutes.
- **Standard normal table** ($\Phi(z) = P(Z \leq z)$ for $Z \sim N(0, 1)$), needed only where a probability is asked:

z	0.50	1.00	1.28	1.50	1.645	1.96	2.00	2.50	2.58	3.00
$\Phi(z)$	0.6915	0.8413	0.8997	0.9332	0.9500	0.9750	0.9772	0.9938	0.9951	0.9987

Use $P(Z > z) = 1 - \Phi(z)$ and, by symmetry, $\Phi(-z) = 1 - \Phi(z)$.

- **Question mix:** Q1 derivation (OLS FOCs and residual identities); Q2 regression-output interpretation (prediction, change in prediction, units); Q3 large-sample sampling distribution of $\hat{\beta}_1$ (standardising, probability calculations, the role of $SE(\hat{\beta}_1)$); Q4 binary regressor (group means, an OLS-with-a-dummy proof, Least Squares Assumption 1 in this setting); Q5 conceptual (causality, Least Squares Assumption 1, selection bias, Type I/II trade-off).

Instructions

Answer **all** questions. Show all working clearly; answers without justification may receive zero marks.

Question 1**(10 Marks)**

Consider the simple linear regression model

$$Y_i = \beta_0 + \beta_1 X_i + u_i, \quad i = 1, \dots, n.$$

For candidate values b_0, b_1 , define the sum of squared errors

$$S(b_0, b_1) = \sum_{i=1}^n (Y_i - b_0 - b_1 X_i)^2.$$

The OLS estimators $\hat{\beta}_0, \hat{\beta}_1$ are the values of b_0, b_1 that minimize $S(b_0, b_1)$.

- (a) (4 marks) Derive the two first-order conditions for this minimization problem, and show that they can be written as

$$\sum_{i=1}^n (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i) = 0, \quad \sum_{i=1}^n (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i) X_i = 0.$$

- (b) (3 marks) Using part (a), show that the OLS residuals $\hat{u}_i = Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i$ satisfy

$$\sum_{i=1}^n \hat{u}_i = 0, \quad \sum_{i=1}^n \hat{u}_i X_i = 0.$$

- (c) (3 marks) Using part (b), show that $\sum_{i=1}^n \hat{u}_i \hat{Y}_i = 0$, where $\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_i$ is the OLS predicted value.

Question 2**(10 Marks)**

A researcher studying the Singapore private rental market collects data on $n = 150$ HDB flat rental listings and regresses monthly rent (in SGD) on floor area (in square metres, sqm). The estimated OLS regression line is

$$\widehat{\text{Rent}}_i = 380 + 24.5 \text{Area}_i.$$

- (a) (3 marks) A 90 sqm flat is listed for rent. What is the regression's predicted monthly rent for this flat?
- (b) (3 marks) A tenant moves from a 70 sqm flat to a 95 sqm flat. What is the regression's predicted change in monthly rent?
- (c) (4 marks) Interpret the slope coefficient, including units. Is the intercept economically meaningful in this context? Explain briefly.

Question 3**(10 Marks)**

A team of researchers studies $n = 250$ Singapore tech start-ups, regressing monthly revenue growth (%) on monthly R&D spending (SGD '000s). Based on the large sample, they treat the OLS estimator $\hat{\beta}_1$ as following its large-sample sampling distribution

$$\hat{\beta}_1 \sim N(\beta_1, \sigma_{\hat{\beta}_1}^2),$$

with population slope $\beta_1 = 2.0$ and standard deviation of the sampling distribution $\sigma_{\hat{\beta}_1} = \text{SE}(\hat{\beta}_1) = 0.5$.

- (a) (2 marks) Explain, in words, why $\hat{\beta}_1$ – computed from this one sample of 250 start-ups – is itself a random variable, even though the population slope β_1 is a fixed (if unknown) number.
- (b) (3 marks) Using the large-sample result $\frac{\hat{\beta}_1 - \beta_1}{\sigma_{\hat{\beta}_1}} \sim N(0, 1)$, standardise and use the standard normal table in the Overview to compute $P(\hat{\beta}_1 > 2.75)$.
- (c) (3 marks) Compute $P(|\hat{\beta}_1 - \beta_1| < 0.98)$, i.e. the probability that the estimator falls within 0.98 of the true population slope. Use the standard normal table in the Overview.
- (d) (2 marks) Explain what $\text{SE}(\hat{\beta}_1)$ measures, and give two reasons – referring to the formula for $\sigma_{\hat{\beta}_1}^2$ – why it becomes *smaller* as (i) the sample size n increases, and (ii) the sample spread of the regressor X increases.

Question 4**(10 Marks)**

A study of $n = 200$ food-delivery riders in Singapore regresses monthly income (in SGD) on D_i , a dummy variable equal to 1 if the rider owns their vehicle and 0 if the rider rents it ($n_0 = 120$ renters, $n_1 = 80$ owners). The estimated regression is

$$\widehat{\text{Income}}_i = 1450 + 320 D_i, \quad \text{SE}(\hat{\beta}_1) = 95.$$

- (a) (3 marks) State the average monthly income predicted by the regression for renters and for owners.
- (b) (2 marks) Give an economic interpretation of $\hat{\beta}_1 = 320$.
- (c) (3 marks) Prove, using the OLS normal equations from Question 1 applied to this binary regressor $D_i \in \{0, 1\}$, that in general $\hat{\beta}_0 = \bar{Y}_0$ (the sample mean of income among renters) and $\hat{\beta}_1 = \bar{Y}_1 - \bar{Y}_0$ (the difference in sample mean income between owners and renters), where $\bar{Y}_0, \bar{Y}_1$ are the group sample means and n_0, n_1 are the group sizes.
- (d) (2 marks) State what Least Squares Assumption 1 ($E[u_i | D_i] = 0$) requires in this dummy-regressor setting, give one specific unobserved factor that could plausibly make it fail here, and state the likely direction of the resulting bias in $\hat{\beta}_1$.

Question 5**(10 Marks)**

- (a) (4 marks) A researcher regresses PSLE (Primary School Leaving Examination) score on weekly hours of private tuition, using observational data on Singapore primary-six pupils. She finds a strong positive coefficient on tuition hours and concludes that private tuition *causes* higher PSLE scores. Using Least Squares Assumption 1 ($E[u_i|X_i] = 0$) and the concept of selection bias, explain why this conclusion may not be valid. In your answer, state a specific unobserved factor that could be correlated with tuition hours, and predict the likely direction of the resulting bias.
- (b) (3 marks) Suggest one research design that would let the researcher obtain a more credible estimate of the causal effect of tuition on PSLE score, and briefly explain why it addresses the problem identified in part (a).
- (c) (3 marks) A bank's automated loan-approval system flags each application as "high risk" (reject) or "safe" (approve) based on a credit-score cutoff. Define, in the context of this bank's decision, what a Type I error and a Type II error would mean, and explain the trade-off the bank faces if it raises the credit-score cutoff required for approval.